PES UNIVERSITY, Bangalore

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Department of Computer Science & Engineering

UE25MA242A: Mathematical Foundation for AI & Data Science

Mini-Project Problem Statements

Application Problem	Problem Statement
1	Basic operations with matrices in Pythton.
2	Matrix operations and image manipulation
3	Matrix multiplication, inversion, and photo filters
4	Solving linear systems in Python
5	Systems of linear equations and college football team ranking (with an example of the Big 12
6	Convolution, inner product, and image processing revisited
7	Norms, angles, and your movie choices
8	Interpolation, extrapolation, and climate change
9	Orthogonal matrices and 3D graphics
10	Discrete dynamical systems, linear transformations of the plane, and the Chaos Game
11	Projections, eigenvectors, Principal Component Analysis, and face recognition algorithms
12	Matrix eigenvalues and the Google's PageRank algorithm
13	Social networks, clustering, and eigenvalue problems
14	Singular Value Decomposition and image compression